

Codes and World War II

America's involvement in the Second World War was the result of an intercepted cipher. Known as the 'Zimmermann Telegram', this cable from the German Foreign Office was delivered through Washington to its ambassador Heinrich von Eckardt in Mexico. This was an invitation for Mexico to ally with Germany and in return, they were promised the chance to invade the United States of America. This was thwarted as the cipher was intercepted, and played a massive role in getting America involved in World War II.

Overcoming Enigma

During World War II, information could win or lose a battle and cryptanalysis was an integral part of it. The Nazis relied on the Enigma machine which was used for cryptography to secure intelligence which was vital. By this time, mechanical and electromechanical ciphers were widely utilized. After the telegraph, the advent of the radio changed the game and further developments were made when the rotor system of cryptography was used. Radio completely changed the landscape and French radio stations intercepted most German radio transmissions, though the latter used a double columnar transposition called Ubchi, which was not very strong.

Germany utilized cryptography immensely in all its forms. The Polish Cipher Bureau helped with decoding the detailed structure of the Enigma using mathematics while documentation was given by French military intelligence. This was perhaps the biggest breakthroughs in the history of cryptanalysis. The British Cryptanalysis Department was made up of chess masters and mathematicians, who helped in decoding of the Enigma's encryption. But these officers were not allowed to openly show their achievement at having broken the ciphers fearing the Germany would claim that Britain had not waged a fair war.

Winning wars

The US Navy was able to break into the codes of the Japanese Navy and this helped them gain an edge in the Battle of Midway, ensuring victory. Later on, the highly advanced Japanese diplomatic cipher system with an electromagnetic stepping switch machine, nicknamed "Purple" by the Americans, was broken. The Americans termed the intelligence emerging from Purple as Magic.